Jmal C's Digital Literacy Class

Chapter Summaries · IC3 Computing Fundamentals, Chapters 1-7

CHAPTER 1 OF 7 · ABOUT 9 MIN READ

🪟 Chapter 1: Operating Systems

The operating system (OS) is the boss program on every computer, tablet, and phone. It runs the hardware, starts your apps, and gives you a screen to work on. This chapter is about Windows 10 first, plus what is the same on Macs, Linux, and phones.

What an operating system does

- Controls the hardware: memory, storage, screen, keyboard, printer, network.
- Runs apps and keeps them from crashing into each other.
- Gives you the interface: the desktop, windows, icons, and menus (a graphical user interface, or GUI). Very old systems used typed commands only.
- Keeps each person's files and settings separate with user accounts.

Popular operating systems

OS	Made by	Where you see it	Open or closed?
Windows (10, 11)	Microsoft	Most PCs and laptops	Proprietary (closed)
macOS	Apple	Mac computers only	Proprietary
Linux (Ubuntu and others)	Community	Servers, some desktops	Open source
UNIX	Several companies	Big servers; macOS grew from it	Mostly proprietary
ChromeOS	Google	Chromebooks	Built on Linux
iOS / iPadOS	Apple	iPhone, iPad	Proprietary
Android	Google	Most other phones and tablets	Open source base

- Proprietary: one company owns the code. You buy a license and agree to the EULA (end user license agreement).
- Open source: the code is public. Anyone can study it, change it, and share it, often for free.
- Versions are releases over time (Windows 10, Windows 11). Editions are packages of one version for different buyers (Home, Pro, Enterprise).

User accounts and profiles

Account type	What it can do
Administrator	Change settings for everyone, install or remove programs and hardware, create and manage other accounts
Standard user	Use programs and change only your own settings. Asks an admin to approve big changes
Guest / child accounts	Limited use, often with family safety rules

- A Microsoft account (email sign-in) syncs your settings across PCs. A local account lives on one PC only.
- Your user profile holds your desktop, documents folder, wallpaper, and app settings. Each person on the PC gets their own profile.
- Sign in with a password, PIN, picture password, or fingerprint/face (Windows Hello).

Turning it on and off

Option	What happens	Use it when
Shut down	Closes everything and powers off	Done for the day, moving the computer, or adding hardware
Restart	Shuts down, then starts again	After updates or installs, or when things act strange
Sleep	Keeps your work in RAM, uses a little power, wakes fast	Short breaks
Hibernate	Saves your work to the drive, then powers off	Long breaks on a laptop, to save battery
Lock	Hides your screen; your apps keep running	Stepping away for a minute
Sign out	Closes your apps and leaves the PC on for someone else	Shared or public computers
Switch user	Leaves you signed in while someone else signs in	Shared family PC

- Always shut down from the Start menu Power button. Holding the power button is only for a frozen computer, because you can lose unsaved work.

Getting around the Windows desktop

Part	What it is for
Start button and Start menu	Open apps, search, settings, and the Power button

Part	What it is for
Search box (Cortana in Windows 10)	Type to find apps, files, and settings
Taskbar	Strip at the bottom: pinned apps and open windows
Task View (Windows key + Tab)	See all open windows and add virtual desktops
Notification area	Clock, volume, Wi-Fi, battery icons at the right end of the taskbar
Action Center (Windows 10)	Notifications and quick settings like Wi-Fi and Airplane mode
Dialog box	A small window that asks you to choose options, then OK or Cancel

- Mouse: click to select, double-click to open, right-click for a shortcut menu, drag to move.
- Keyboard: Windows key opens Start; Ctrl+C copy, Ctrl+V paste, Ctrl+Z undo; Alt+Tab switches windows.
- Touch: tap = click, press and hold = right-click, swipe to scroll, pinch or spread to zoom.

Global settings vs. profile settings

Global (whole computer, every user)	Profile-specific (just you)
Screen resolution	Desktop background (wallpaper)
Date and time	Theme colors and lock screen picture
Power and sleep plans	Start menu and taskbar pins
Installing hardware and programs	Screen saver, mouse and sound preferences

- Settings app: the modern place for most options (Start > gear icon).
- Control Panel: the older, classic tool. Still has some options the Settings app doesn't.
- Global changes usually need an administrator.

Mobile operating systems

- Power button: press to wake or lock; press and hold to power off or restart.
- Lock screen: shows time and alerts; swipe and enter your PIN, pattern, fingerprint, or face to unlock.
- Home screen: app icons and widgets. Swipe between pages.
- Settings app: Wi-Fi, Bluetooth, notifications, display, battery, updates.
- Virtual assistants answer voice questions: Siri (Apple), Google Assistant (Android), Cortana (Windows 10).

Operating system updates

Windows update type	What it contains
Important	Security and critical fixes. Install right away
Recommended	Bug fixes and improved features
Optional	Extra drivers or trial software you pick yourself

- Windows 10 and 11 download and install updates automatically. You can pause them for a while or set active hours so the PC won't restart while you work.
- Check by hand: Settings > Update & Security > Windows Update > Check for updates (Windows 11: Settings > Windows Update).
- Good: you get security fixes fast. Bad: a surprise restart or, rarely, an update that breaks a driver.
- Phones: Settings > Software update (Android) or Settings > General > Software Update (iPhone). Plug in and use Wi-Fi first.

🔤 Key vocabulary

Operating system (OS)
Main software that runs the device and your apps

GUI
Graphical user interface: windows, icons, menus

Proprietary
Owned by one company; code is private

Open source
Code is public and can be changed and shared

EULA
The license terms you accept before using software

Administrator
Account that can change settings for everyone

Standard user
Everyday account that changes only its own settings

User profile
Your personal files and settings on a PC

Hibernate
Save work to the drive and power off

Sleep
Low-power pause that keeps work in memory

Taskbar
Bar at the bottom with pinned and open apps

Task View
Shows open windows and virtual desktops

Control Panel
Older Windows tool for system settings

Patch / update
A fix or improvement from the software maker

✅ Know for the test

- Administrator vs. standard user: only admins install programs and change settings for everyone.
- Sleep keeps work in RAM; hibernate saves it to the drive and turns off. Lock keeps apps running; sign out closes them.
- Wallpaper and theme are profile settings. Resolution, date and time, and power plans are global.
- Right-click (or press and hold on touch) opens a shortcut menu.
- Important updates are security fixes. Windows 10/11 installs updates automatically.
- Windows and macOS are proprietary. Linux is open source. Android is built on open source.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. You will be away from your laptop for a weekend and want your open files waiting when you come back. Which power option?

Answer: Hibernate. It saves your work to the drive and powers off, so the battery isn't drained.

Q2. Which account type can install a new program for everyone on the PC?

Answer: An administrator account.

Q3. You change the desktop background. Does your sister's account on the same PC change too?

Answer: No. The background is a profile-specific setting.

Q4. What does pressing the Windows key + Tab open?

Answer: Task View, where you see open windows and can make a new virtual desktop.

Q5. Name the three kinds of Windows updates.

Answer: Important, recommended, and optional.

CHAPTER 2 OF 7 · ABOUT 8 MIN READ

Chapter 2: Hardware

Hardware is the physical stuff you can touch. This chapter explains how hardware is measured, the difference between memory and storage, the main kinds of computers, input devices, power plans, ports, and wireless connections.

Drivers, firmware, and platforms

Term	Plain meaning	Example
Device driver	Software inside the OS that lets it talk to one piece of hardware	Printer driver, graphics driver
Firmware	A small program stored on the device's own chip that runs the hardware at the most basic level	BIOS/UEFI on a PC, phone or router firmware
Platform	The hardware plus OS combination that apps are built for	Windows PC, Mac, Android phone, iPhone

- Update drivers after a big OS upgrade or if a device stops working right. Windows Update handles most of them.
- An app built for one platform usually won't run on another (an iPhone app won't install on Windows).

Units of measure

Unit	Short	About how much
Bit	b	One 0 or 1, the smallest piece of data
Byte	B	8 bits, about one typed character
Kilobyte	KB	About 1,000 bytes (1,024). A short text file
Megabyte	MB	About 1,000 KB. A photo or a song
Gigabyte	GB	About 1,000 MB. A movie; phone and RAM sizes
Terabyte	TB	About 1,000 GB. A big hard drive
Hertz / GHz	Hz	Cycles per second. CPU speed is in gigahertz (billions per second)
Bits per second	Kbps, Mbps, Gbps	Network and internet speed (bandwidth)

- Capital B = bytes (storage). Small b = bits (speed). 100 Mbps is megabits per second, not megabytes.

Memory vs. storage

	Memory (RAM)	Storage (drive)
Job	Short-term workspace for open apps	Long-term home for files and programs
When power is off	Erased (volatile)	Kept (non-volatile)
Typical size	4 to 32 GB	128 GB to several TB
Too little causes	Slow, freezing, crashes with many apps open	Can't save or install; device out of space
Fix	Close apps and tabs, restart, add RAM	Delete or move files, empty the Recycle Bin, add a drive or use the cloud

- Storage devices: hard disk drive (HDD, spinning, cheap and large), solid state drive (SSD, fast, no moving parts), USB flash drive, memory card, external drive, CD/DVD/Blu-ray.
- Local storage is inside or plugged into your device. Remote storage is on another computer: a network drive or the cloud.

Types of computers

Type	Best for	Trade-offs
Server	Sends files, websites, email, or apps to other computers on a network	Powerful and runs all the time; usually managed by IT
Desktop	Power, big screens, easy to upgrade	Not portable
Laptop	Work anywhere with a built-in screen, keyboard, and battery	Harder to upgrade; battery limits
Chromebook	Low cost, web apps, cloud storage (ChromeOS)	Small local storage; needs internet for most things
Tablet	Touch, reading, video, light work	Typing and big programs are harder
Smartphone	Calls, texts, apps, camera, GPS	Small screen; battery and data limits

Keyboards, mice, and touch

- Keyboard keys: letters and numbers, function keys (F1 to F12; F1 is often Help), arrow keys, Ctrl, Alt, Shift, and the Windows key.
- Keyboards connect by USB cable or wirelessly (Bluetooth or a small USB receiver).
- Pointing devices: mouse, touchpad on laptops, and stylus (a pen for touch screens).
- Touch screens are easy to learn and need no extra parts, but they are less precise than a mouse and smudge easily.

- Smartphone hardware: touch screen, cameras, microphone, speaker, GPS, sensors, battery, and a SIM card that identifies you to the cell carrier.

Power plans and battery life

Windows plan	What it does
Balanced	Full speed when needed, saves energy when idle (default)
Power saver	Dims the screen and slows down to stretch battery
High performance	Maximum speed, uses the most power

- Find it: Control Panel > Power Options, or Settings > System > Power & sleep (Windows 11: Power & battery).
- Battery killers: bright screen, Wi-Fi and Bluetooth left on, many apps running, video and games, an old battery.

Ports and peripherals

Port	Used for
USB (Type-A, Type-C)	Almost everything: mouse, keyboard, printer, flash drive, phone charging
HDMI	Digital video and sound to a TV or monitor
DisplayPort / DVI	Digital video to monitors
VGA	Older analog video (blue plug)
Ethernet (RJ-45)	Wired network cable
Audio jacks (3.5 mm)	Headphones, speakers, microphone
PS/2, serial, parallel	Old ports for keyboards, mice, and printers

- Peripherals are devices you add on: monitor, printer, scanner, speakers, webcam, external drive, digital camera.
- A second monitor can mirror your screen (duplicate) or give you more space (extend). Press Windows key + P to choose.

Wireless connections

Technology	Range	Typical use
Wi-Fi	A home or building	Internet and network access
Bluetooth	About 30 feet (10 m)	Earbuds, keyboards, mice, car audio. Devices must be paired
Infrared (IR)	Short; needs a clear line of sight	TV remotes, some older devices

Technology	Range	Typical use
Cellular	Miles, from towers	Phones and cellular tablets

🔤 Key vocabulary

Driver
Software that lets the OS control a device

Firmware
Built-in program on a device's chip

Byte
8 bits

RAM
Fast temporary memory, erased when off

Volatile
Loses data without power

SSD
Fast drive with no moving parts

HDD
Spinning hard disk drive

CPU
Processor; does the computer's thinking

GHz
Billions of cycles per second (CPU speed)

Peripheral
An add-on device like a printer or webcam

Stylus
Pen for a touch screen

SIM card
Chip that links a phone to a carrier

Pairing
Linking two Bluetooth devices the first time

Server
Computer that provides services to others on a network

✅ Know for the test

- RAM is temporary and volatile; storage is permanent. More RAM helps multitasking; more storage holds more files.
- Order: bit, byte, KB, MB, GB, TB. Storage uses bytes; speed uses bits per second.
- A server answers requests from other computers.
- A driver lives in the OS; firmware lives on the device.
- HDMI carries video and sound; VGA is old analog video; RJ-45 is Ethernet.
- Bluetooth is short range and must be paired; infrared needs line of sight.
- Power saver stretches battery; high performance drains it.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. Your laptop freezes when you have 30 browser tabs open. What hardware upgrade helps most?

Answer: More RAM (memory).

Q2. Which is bigger, 500 MB or 2 GB?

Answer: 2 GB. One GB is about 1,000 MB.

Q3. Which port sends both video and sound to a TV?

Answer: HDMI.

Q4. Wireless earbuds need to be connected to your phone the first time. What is that called?

Answer: Pairing (Bluetooth).

Q5. What is the difference between a driver and firmware?

Answer: A driver is software in the OS that talks to a device. Firmware is built into the device's own chip.

CHAPTER 3 OF 7 · ABOUT 9 MIN READ

📶 Chapter 3: Networks and Mobile Devices

A network is two or more devices linked so they can share data, printers, and an internet connection. This chapter covers how home and business networks work, how you get internet service, wired and wireless connections, cell service, landlines, and voice mail.

Network basics

- Networks let people share files, printers, and one internet connection, and let you stream (watch while it arrives) or download (save a full copy first).
- TCP/IP is the set of rules (protocols) every device on the internet uses to send data.
- LAN (local area network): one home, office, or school. WAN (wide area network): links places far apart. The internet is the biggest WAN.
- Businesses also use intranets (private, for employees) and extranets (shared with partners).

IP addresses

Idea	What to know
IP address	The number that identifies each device on a network
IPv4	Four numbers 0 to 255 with dots, like 192.168.1.10
IPv6	Newer and much longer, with letters and colons; made because IPv4 addresses ran out
Private address	Used inside your home or office LAN (often starts 192.168 or 10.)
Public address	Given to your router by your ISP; what websites see
Find yours	Settings > Network details, or type ipconfig in Command Prompt

- Your router hands out private addresses to each device automatically (DHCP). Two devices on one network can't share the same address.

Getting internet service

Service	Comes over	Notes
Dial-up	Phone line, by calling in	Very slow and mostly gone
DSL	Phone company's copper lines	Speed depends on distance from the phone office
Cable	TV company's coaxial cable	Fast, but shared with neighbors, so it slows at busy times
Fiber (like FiOS)	Glass strands carrying light	Fastest; not available everywhere

Service	Comes over	Notes
Satellite	Dish pointed at a satellite	Works in rural areas; slower response and data limits
Cellular (4G, 5G) and hotspots	Cell towers	Works on the go; usage counts against your data plan

- Broadband means always-on, high-speed internet (everything here except dial-up).
- An ISP (internet service provider) sells you the connection. Ads say speeds are "up to" a number because real speed is usually lower.
- Download speed (data coming to you) is usually much faster than upload speed (data you send).
- Run a speed test website to check what you really get.

Home network hardware

Device	Job
Modem	Translates the ISP's signal so your network can use it
Router	Shares one internet connection with all your devices and directs traffic between networks
Switch	Adds more wired Ethernet ports to a LAN
Wireless access point	Lets Wi-Fi devices join the network (most home routers have one built in)
Network interface card (NIC)	The part in each device that connects it to a network, wired or wireless

Wired (Ethernet) vs. wireless (Wi-Fi)

	Wired Ethernet	Wi-Fi
Speed and reliability	Faster and steadier	Can slow down with distance, walls, and interference
Security	Harder to break into; someone needs a cable	Must be protected with a password and encryption
Convenience	Stuck where the cable reaches	Move around freely; works for phones and tablets
Connector	RJ-45 cable (Cat5e, Cat6)	Radio waves (2.4 GHz and 5 GHz bands)

- Wi-Fi standards are named 802.11 (a, b, g, n, ac, and newer). Newer ones are faster.
- 2.4 GHz reaches farther but is slower and more crowded. 5 GHz is faster but shorter range.
- Secure Wi-Fi uses WPA2 or WPA3 encryption and a strong password. WEP is old and weak. Change the router's default admin password.

- WPS lets you connect by pressing a button on the router, but it is less secure.
- Join Wi-Fi: click the network icon, pick the network name, enter the password, Connect.
- Shared or network printers can be added from Settings > Devices > Printers & scanners.

Cellular networks and service

- Phones talk by radio to cell towers. Each tower covers an area (a cell) and hands calls off as you move.
- Carriers (Verizon, AT&T, T-Mobile and others) own the networks. Coverage maps show where service is strong.
- Cell service is separate from your home internet; the carrier, not an ISP, bills you for calls, texts, and data.
- Contract plans: lower phone price, but an early termination fee if you leave. Prepaid plans: pay first, no contract, no credit check.
- Plans bundle minutes, texts, and data. Add-ons include hotspot use, international calling, and insurance.
- Roaming means using another carrier's network outside your area, which can cost extra.

Cellular devices

Device	Can do	Limits
Smartphone	Calls, texts, apps, internet, camera, GPS	Battery, screen size, data limits
Basic cell phone	Calls and texts	Few or no apps
Cellular tablet	Internet anywhere with a data plan	May not make regular phone calls; needs its own plan
Wi-Fi-only tablet	Internet where Wi-Fi exists	No connection away from Wi-Fi

- Use Wi-Fi when you can to save mobile data. Turn off mobile data in Settings to avoid charges.

Landlines and business phones

Landline good points	Landline weak points
Very reliable with clear sound	Stuck in one place
A corded phone works during a power outage (the line powers it)	Monthly cost even if you rarely use it
911 knows your address	No texting, apps, or internet

- VoIP phones send calls over the internet. They stop working if the power or internet goes out.

- Businesses use a PBX, a private phone system with extensions, transfers, and often dialing 9 for an outside line.

Voice mail

- Set it up the first time you call your voice mail: create a PIN, record your name, record a greeting.
- A professional greeting: your name, "please leave your name, number, and a short message", and when you'll call back.
- Check messages often. A full mailbox means callers can't leave a message, so delete old ones.
- Leaving a message: say your name and number slowly, the reason for the call, and repeat the number at the end.
- Many phones show visual voice mail, a list you can tap and play in any order.

🔤 Key vocabulary

LAN
Network in one building or home

WAN
Network that covers a large area

TCP/IP
Rules devices use to talk on the internet

IP address
Number that identifies a device on a network

ISP
Company that sells internet service

Bandwidth
How much data can move per second

Modem
Converts the ISP's signal for your network

Router
Shares internet and directs traffic between networks

NIC
Network adapter in a device

SSID
The name of a Wi-Fi network

WPA2 / WPA3
Strong Wi-Fi encryption

Hotspot
A phone or device sharing its cell data over Wi-Fi

Carrier
Company that runs a cell network

VoIP
Phone calls over the internet

PBX
Private business phone system

✅ Know for the test

- Modem connects to the ISP; router shares the connection and links networks.
- Wired is faster, steadier, and more secure; wireless is more convenient.
- Cable is shared with neighbors; DSL uses phone lines; satellite is for rural areas.
- Mbps measures speed. Download is usually faster than upload.

- IPv4 has 4 numbers with dots. Every device on a network needs a unique IP address.
- A corded landline works in a power outage; VoIP does not.
- Professional voice mail greeting: name, request for caller's name and number, call-back promise.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. Which device shares one internet connection with every device in your home?

Answer: The router.

Q2. Your neighbor's internet slows down every evening when everyone gets home. Which service do they probably have?

Answer: Cable, because it is shared.

Q3. Is 192.168.1.25 an IPv4 or IPv6 address?

Answer: IPv4 (four numbers separated by dots).

Q4. The power goes out. Which phone still works: a corded landline or a VoIP phone?

Answer: The corded landline.

Q5. What should a professional voice mail greeting include?

Answer: Your name, a request for the caller's name, number, and message, and that you will call back.

CHAPTER 4 OF 7 · ABOUT 8 MIN READ

📁 Chapter 4: File Management

File management means keeping your work organized so you can find it, share it, and not lose it. This chapter covers drives, folders, paths, permissions, File Explorer, file types, copying and moving, searching, the Recycle Bin, and sharing files.

Drives, folders, and paths

Item	What it is
C: drive	The main internal drive with Windows and your programs
D:, E:, and up	Other drives: second hard drive, DVD, USB flash drive, memory card
Mapped network drive	A shared folder on another computer that shows up with its own letter
Folder and subfolder	Containers for files; a folder inside a folder is a subfolder
Path	The full address of a file, like C:\Users\Ana\Documents\Resume.docx

- The folder structure is often called a directory tree. Paths use backslashes (\) in Windows and forward slashes (/) on the web and on Macs.
- When you connect a smartphone, tablet, or camera by USB, it appears in File Explorer under This PC so you can copy files from it.

Permissions

Permission	Lets a user
Read	Open and view a file but not change it
Write / Modify	Change and save the file
Full control	Do everything, including change permissions and delete

- Permissions protect files on shared PCs and networks. An administrator sets them.
- Local files are on your device; remote files are on a network or the cloud and may have different rules.

Using File Explorer

- Open it with the folder icon on the taskbar or Windows key + E.
- Navigation pane (left): Quick Access, OneDrive, This PC, Network. Address bar (top): shows the path. Ribbon or toolbar: commands.
- Views: icons, list, details (shows date, type, size), tiles. Sort by clicking a column heading like Date modified.

- Window controls: minimize (hide to taskbar), maximize (full screen), restore, close. Drag the title bar to move; drag an edge or corner to resize. Use scroll bars to see more.

File types and extensions

Kind	Common extensions
Documents	.docx (Word), .xlsx (Excel), .pptx (PowerPoint), .pdf, .txt, .rtf
Pictures	.jpg / .jpeg, .png, .gif, .bmp, .tif
Audio	.mp3, .wav, .m4a, .wma
Video	.mp4, .mov, .avi, .wmv
Programs (executable)	.exe, .msi (be careful opening these)
Compressed (archive)	.zip, .rar, .7z

- The extension tells Windows which app opens the file. Windows hides extensions by default; turn them on in File Explorer: View > File name extensions.
- Don't change an extension when renaming. The file may stop opening.

Working with files and folders

Task	How
New folder	Right-click empty space > New > Folder, or Ctrl+Shift+N
Rename	Select it and press F2, or right-click > Rename
Copy	Ctrl+C then Ctrl+V. The original stays
Move	Ctrl+X then Ctrl+V, or drag within the same drive
Select several	Ctrl+click picks one at a time; Shift+click picks a range; Ctrl+A picks all
Shortcut	Right-click > Send to > Desktop (create shortcut). Deleting a shortcut doesn't delete the file

- Dragging to a folder on the same drive moves the file. Dragging to a different drive (like a USB stick) copies it.
- Folder Options lets you show hidden files, open items with a single click, and more.

Searching and the Recycle Bin

- Search from the taskbar box for apps, files, and settings, or use the search box in File Explorer to look inside the current folder.
- Narrow results with filters such as date modified, kind (type), and size, or search for an extension like .pdf.

- Deleted files from your internal drive go to the Recycle Bin. Open it and choose Restore to put a file back where it was.
- Emptying the Recycle Bin removes the files for good. Files deleted from USB drives and network folders skip the Recycle Bin.
- Shift+Delete deletes permanently without using the Recycle Bin.

Default locations

Item	Usually saved to
Downloads from the web	Downloads folder
Documents	Documents folder (or OneDrive)
Photos and scans	Pictures folder
Music and videos	Music and Videos folders

- Phone photos can be copied by USB cable, uploaded to the cloud (Google Photos, iCloud, OneDrive), or sent with Bluetooth or a memory card.
- Media apps (Photos, media players) sort pictures, music, and video into albums and libraries.

Sharing files

Method	Good for	Watch out for
USB flash drive or external drive	Big files, no internet needed	Easy to lose; can carry malware
CD / DVD	Older archives	Small capacity, slow, few new PCs have drives
Public or shared folder	People on the same PC or network	Set permissions carefully
Network share	Office teams	Needs network access and permissions
Email attachment	Small files	Size limits around 20 to 25 MB; some file types blocked
Cloud link	Large files and teamwork	Needs internet; choose view or edit access

- Zipping (compressing) bundles many files into one smaller .zip file. Right-click > Send to > Compressed (zipped) folder. To use the files, right-click > Extract All.
- Zip before emailing a group of files or when a file is a little too large. Photos and videos don't shrink much because they are already compressed.

🔤 Key vocabulary

Path
Full address of a file or folder

Directory
Another word for a folder structure

Subfolder
A folder inside another folder

Extension
Letters after the dot that show the file type

Executable
A file that runs a program (.exe)

Shortcut
A link that opens a file or folder

Recycle Bin
Holding area for deleted files

Permissions
Rules for who can read or change a file

Mapped drive
Network folder shown as a drive letter

Zip / compress
Bundle and shrink files into one

Extract
Unpack files from a zip

Default location
Where a type of file is saved unless you choose

✅ Know for the test

- Copy keeps the original; move doesn't. Same-drive drag moves; different-drive drag copies.
- Ctrl+click for separate files, Shift+click for a range.
- Recycle Bin restores files from the internal drive only; emptying it is permanent.
- Know common extensions: .docx, .xlsx, .pptx, .pdf, .jpg, .png, .mp3, .mp4, .exe, .zip.
- Web downloads go to the Downloads folder.
- Read permission = view only; write = change; full control = everything.
- Zip to send many files as one; extract to use them.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. Write the path for Notes.txt in a folder named Class inside Documents on the C: drive.

Answer: C:\Users\(your name)\Documents\Class\Notes.txt, or simply Documents\Class\Notes.txt.

Q2. You delete a file from a USB flash drive. Can you get it back from the Recycle Bin?

Answer: Usually no. Files deleted from flash drives are removed right away.

Q3. Which extension is a compressed folder?

Answer: .zip

Q4. Where does a file you download from a website usually go?

Answer: The Downloads folder.

Q5. How do you select files 1, 4, and 7 in a list, but not the ones in between?

Answer: Hold Ctrl and click each one.

CHAPTER 5 OF 7 · ABOUT 7 MIN READ

💿 Chapter 5: Software

Software is the programs that let you do things on a device: write, calculate, browse, chat, and more. This chapter covers where to get software, checking requirements, installing, removing, repairing, updating, setting preferences, app stores, and messaging apps.

Getting software

Way to get it	What to know
App store (Microsoft Store, Apple App Store, Google Play)	Safest; apps are checked and update themselves
Software company's website	Download from the official site only
Disc or USB	Older way; still used for some programs
Subscription (Microsoft 365, Adobe)	Pay monthly or yearly; updates included while you pay
One-time purchase	Pay once for that version
Freeware / open source	Free to use; open source also shares its code
Free trial	Works for a limited time, then you must buy it

- Platform matters: get the version made for your OS (Windows, macOS, Android, iOS).
- Check system requirements first: OS version, processor, RAM, free storage, and sometimes graphics or internet.

Installing, uninstalling, repairing, updating

Task	Where in Windows	Why
Install	Run the installer (setup.exe or .msi) or click Install in the Store	Add a new program
Uninstall	Settings > Apps > (app) > Uninstall, or Control Panel > Programs > Uninstall a program	Remove it fully and free space
Repair	Settings > Apps > (app) > Modify / Advanced options > Repair	Fix a program that crashes or won't start
Reinstall	Uninstall, then install again	When repair doesn't work
Update	Inside the app (Help > Check for updates), the Store, or Windows Update	Security fixes and new features

- User Account Control (UAC) pops up to ask "Do you want to allow this app to make changes?" A standard user needs an admin password.

- You must accept the EULA (license agreement) to install. Some programs also ask you to register or enter a product key.
- Don't remove a program by deleting its desktop shortcut. That only removes the icon.

Configuring software

You want to	Where to change it
Change options for one app (Word, Excel)	Inside the app: File > Options
Add buttons you use a lot (Print, Save As)	Customize the Quick Access Toolbar
Change the starting font, size, or save folder	The app's options (program defaults)
Set how often work is saved for you	AutoRecover or AutoSave settings
Choose which app opens links, photos, or PDFs	Windows Settings > Apps > Default apps

- App settings change only that app. Windows settings change the whole system.

Windows apps and the Microsoft Store

Store task	How
Sign in	Use your Microsoft account (profile icon at the top)
Find an app	Search, then compare ratings, reviews, price, and size
Install	Click Get (free) or the price button, then Install
Remove	Right-click the app on the Start menu > Uninstall
Get it back	Store > Library lists apps you own; reinstall from there

- Sign in to the Store with a Microsoft account to download apps; free and paid apps are listed.
- Search, check ratings, reviews, size, and requirements, then click Get or Install.
- Remove a Store app: right-click it on the Start menu > Uninstall.
- Get a deleted app back: Store > Library shows apps you own so you can reinstall them.
- Sign out of the Store on a shared computer.

Messaging apps

Type	How it works	Limits
SMS (text)	Short text messages through your cell carrier	About 160 characters; text only; uses your plan
MMS	Picture, video, or group messages through your carrier	File size limits; may cost more

Type	How it works	Limits
Chat / IM apps (WhatsApp, iMessage, Messenger, Teams)	Messages over the internet, often with video calls, files, and groups	Need Wi-Fi or data; both people need the app or account
Gmail chat / Google Chat, Skype (retired 2025, now Teams)	Chat from a browser or app with contacts	Must add contacts; need internet

- Presence (status) shows if a contact is available, busy, or away.
- Be careful what you send. Messages and screenshots can be saved and shared.

🔤 Key vocabulary

Application (app)
A program that does a job for you

System requirements
Minimum hardware and OS a program needs

Installer
File that sets up a program

EULA
License terms you accept

UAC
Windows prompt before changes that affect the system

Repair
Fix damaged program files

Patch
Small update that fixes problems

Subscription
Pay over time to keep using software

Freeware
Free software the owner still controls

Quick Access Toolbar
Customizable buttons at the top of Office apps

SMS
Plain text message through a carrier

MMS
Picture or video message through a carrier

Instant messaging
Real-time chat over the internet

✅ Know for the test

- Check system requirements before installing.
- Uninstall from Settings > Apps or Control Panel; deleting a shortcut doesn't remove a program.
- Try Repair before reinstalling a program that crashes.
- UAC asks permission for system changes; standard users need an admin password.
- App settings (like Word's default font) affect only that app.
- SMS is short text through the carrier; MMS adds pictures; chat apps need the internet.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. Before buying a game, what should you check so you know it will run on your PC?

Answer: The system requirements.

Q2. Excel crashes every time it opens. What should you try before reinstalling?

Answer: Run Repair from Settings > Apps (or Control Panel > Programs).

Q3. What is the difference between SMS and MMS?

Answer: SMS is short text only. MMS can include pictures, video, or group messages.

Q4. What pops up in Windows to ask if an app can make changes to your device?

Answer: User Account Control (UAC).

Q5. Where do you change which browser opens links on Windows?

Answer: Settings > Apps > Default apps.

CHAPTER 6 OF 7 · ABOUT 7 MIN READ

☁️ Chapter 6: Cloud Computing

The cloud means using storage and apps that live on a company's servers, reached over the internet, instead of only on your own device. This chapter covers what the cloud is, cloud accounts, Google Drive, OneDrive, iCloud, Dropbox, other cloud apps, and phone notifications.

Cloud basics

Benefits	Drawbacks
Reach your files from any device, anywhere	Needs an internet connection
Easy sharing and working together at the same time	Free space is limited; more costs money
Automatic backup if your device breaks or is lost	You trust the company to keep files safe and private
Apps run in a browser, no install needed	Slow internet means slow work

- To use the cloud you need: an internet connection, a device with a browser or app, and an account with the provider.
- Hosted services: the provider owns and runs the servers; you just sign in.

Cloud accounts

Account	Gives you
Google account (Gmail)	Google Drive, Docs, Sheets, Slides, Photos
Microsoft account (Outlook.com, school or work)	OneDrive, Word, Excel, PowerPoint online
Apple ID	iCloud Drive, Photos, backup for iPhone
Dropbox account	Dropbox storage and sharing

- Manage your account: change your password, turn on 2-step verification, see how much storage you've used, and sign out of devices you don't use.

Working with cloud storage

Action	What it means
Upload	Copy a file from your device to the cloud (New > File upload, or drag it in)
Download	Copy a file from the cloud to your device
Sync	Keep the same files matching on every device automatically

Action	What it means
Share	Send a link or invite people with view or edit access
Organize	Make folders, rename, move, star, and delete in the web page or app

- Deleting a synced file on one device deletes it everywhere. Most services keep it in their own recycle bin for a while.
- OneDrive in File Explorer: a cloud icon means online-only (needs internet); a green check means it's on your device. Choose Always keep on this device for offline use.
- Share settings: Can view (read only) or Can edit. You can share with specific people or anyone with the link. Give the least access needed.

The big cloud storage services

Service	Company	Works best with
OneDrive	Microsoft	Windows and Office; built into File Explorer
Google Drive	Google	Gmail, Chromebooks, Android; Docs, Sheets, Slides
iCloud	Apple	iPhone, iPad, and Mac; backs up photos and settings
Dropbox	Dropbox	Any device; simple sharing of big files

- Web apps (Word, Excel, PowerPoint online; Google Docs, Sheets, Slides) open and edit files right in the browser, and several people can edit at once.
- Each service has a mobile app so you can see, upload, and share files from your phone.

Other cloud apps

Kind	What it does	Examples
SaaS (software as a service)	You use software over the internet instead of installing it, often by subscription	Microsoft 365, Google Workspace
LMS (learning management system)	Online classes: lessons, assignments, quizzes, grades, discussion boards	Blackboard, Canvas, Google Classroom, Moodle
CRM (customer relationship management)	A database of customers, sales, and contacts that a team shares	Salesforce, HubSpot
Discussion board (forum)	Post a question and others reply in threads; not live chat	Class forums, help communities

- CRM helps businesses track every customer contact, follow up on sales, and share info across the team.

Mobile notifications

- Notifications are alerts from apps: banners, sounds, badges (number on the icon), and lock-screen messages.
- Change them in the app's own settings, or for every app in the phone's Settings > Notifications.
- Turn off alerts you don't need to save battery and stay focused. Use Do Not Disturb (Focus on iPhone) for quiet time, with exceptions for important people.
- Calendar reminders and alarms create notifications on purpose.

🔤 Key vocabulary

Cloud computing
Using storage and apps on internet servers

Upload
Send a file from your device to the cloud

Download
Copy a file from the cloud to your device

Sync
Keep files the same across devices

Web app
App that runs in a browser

SaaS
Software used online as a service

LMS
System for online courses

CRM
Shared customer database for a business

Share link
Web address that gives others access to a file

Online-only file
Cloud file not stored on your device

Notification
Alert from an app

Badge
Number on an app icon showing new items

✅ Know for the test

- You need internet and an account to use the cloud.
- Upload = device to cloud; download = cloud to device; sync = keep them matching.
- OneDrive = Microsoft, Google Drive = Google, iCloud = Apple, Dropbox = Dropbox.
- Share with Can view or Can edit; give the least access needed.
- LMS runs online classes; CRM tracks customers; SaaS is software used over the internet.
- Deleting a synced file deletes it on every device.
- Manage notifications in Settings > Notifications or inside each app.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. What three things do you need to use cloud storage?

Answer: Internet, a device with a browser or app, and an account with the provider.

Q2. You want your teacher to read your document but not change it. Which share setting?

Answer: Can view (view only).

Q3. Which cloud service is built into iPhones?

Answer: iCloud.

Q4. What kind of system does a college use for online lessons, assignments, and grades?

Answer: An LMS (learning management system).

Q5. You delete a file from the OneDrive folder on your laptop. What happens on your phone's OneDrive app?

Answer: It is deleted there too, because the folders sync.

CHAPTER 7 OF 7 · ABOUT 9 MIN READ

🔒 Chapter 7: Security and Maintenance

Security keeps your accounts, data, and devices safe. Maintenance keeps them working. This chapter covers passwords, malware, network risks, public computers, scams, antivirus, firewalls, safe shopping, VPNs, backups, restore and reset options, and basic troubleshooting.

Usernames and passwords

- Strong password: long (12+ characters), with uppercase, lowercase, numbers, and symbols, or a passphrase of several random words.
- Avoid names, birthdays, pet names, and common words. Use a different password for each account.
- Never share it. Store passwords in a password manager, not on a sticky note or in an unprotected file.
- Change a password right away if you think someone knows it. Turn on 2-step verification when offered.
- Lock your computer when you walk away (Windows key + L). Sign out on shared computers.

Malware and other threats

Threat	What it does
Virus	Attaches to a file or program and spreads when that file is opened or shared
Worm	Copies itself across networks by itself, with no file needed
Trojan	Pretends to be useful software but opens a back door for attackers
Spyware	Secretly watches what you do and collects your information
Keylogger	Records every key you type, including passwords
Adware	Shows unwanted pop-up ads
Ransomware	Locks your files and demands payment

- Signs of infection: the computer suddenly runs slowly, pop-ups appear, programs crash, settings change, or your friends get strange messages from you.

Network risks

Connection	Risk	Protection
Wired (Ethernet)	Lower; someone needs physical access	Lock up network equipment

Connection	Risk	Protection
Home Wi-Fi	Neighbors or strangers can join or snoop	WPA2 or WPA3, strong Wi-Fi password, change router admin password
Public Wi-Fi (cafe, airport)	Others on the network may see your traffic; fake hotspots	Avoid banking; use HTTPS sites or a VPN; turn off sharing

Public computers

- Don't let the browser save your password, and uncheck "remember me".
- Use a private window (InPrivate or Incognito) if you can.
- Always sign out of every account and close the browser when you finish.
- Clear the history, cookies, temporary files, and form data if you used a normal window.
- Don't save files on the public computer, and watch for people looking over your shoulder.

Social engineering and phishing

- Social engineering tricks people, not computers: a caller pretending to be tech support, a fake boss email, or someone who wants to "verify" your account.
- Phishing is a fake email, text, or website that looks real so you'll type your password or card number. Texts are called smishing; phone calls are vishing.
- Red flags: urgent threats, "click here to avoid closing your account", misspellings, a sender address that doesn't match, and links to odd web addresses.
- Protect yourself: don't click; go to the real website yourself; call the company using a number you trust; never give a PIN or password over the phone; report it.

Protection tools

Tool	What it does	Remember
Antivirus / anti-malware	Scans for and removes viruses and other malware	Keep it on and updated; run full scans (Windows Security / Defender is built in)
Firewall	Filters network traffic and blocks unwanted connections	Hardware firewalls are in routers; software firewalls are on the PC. Keep it on
Monitoring software	Tracks activity (parents, employers)	Can also be used as spyware
HTTPS (padlock)	Encrypts the connection to a website	Look for https:// before entering payment info
VPN	Creates an encrypted tunnel to a trusted network over the internet	Use on public Wi-Fi or to reach a work network from home

- A firewall that's set too strict can block apps you need, so a program may ask you to allow it through.

- Safe online shopping: buy from known stores, be skeptical of deals that seem too good, use a credit card or trusted payment service, and check for HTTPS.

Backup and restore

Tool	What it saves	Use it to
File History (Windows)	Copies of your personal files, saved on a schedule to another drive	Get back an older version or a deleted file
Backup and Restore (Windows 7)	Scheduled backups of files and a system image	Restore files or the whole PC
System Restore point	System files, drivers, and settings (not your documents)	Undo a bad update, driver, or install
System image / repair disc	A full copy of the drive, or a disc to start a broken PC	Rebuild after a drive failure
Cloud backup (iCloud, Google, OneDrive)	Phone photos, contacts, settings	Move to a new phone or recover a lost one

- Back up often and keep a copy somewhere else (another drive, the cloud). The 3-2-1 idea: 3 copies, 2 kinds of media, 1 off-site.
- Encrypt or password-protect backups that hold private information, and keep backup drives in a safe place.

Factory reset

- A factory reset returns a device to how it was when new: your files, apps, and settings are erased.
- Windows 10/11: Settings > Update & Security > Recovery > Reset this PC (Windows 11: System > Recovery). You can keep your files or remove everything.
- Phones: Settings > System > Reset (Android) or Settings > General > Transfer or Reset iPhone.
- Back up first. Reset before you sell or give away a device, or when nothing else fixes a serious problem.

Troubleshooting basics

Problem	Try first
Device won't work	Is it plugged in, charged, and turned on? Check cables and batteries
Program frozen	Wait, then close it with Task Manager (Ctrl+Shift+Esc)
Everything acting strange	Restart the computer
Problem began after a change	Undo it, uninstall the new program, or use System Restore

Problem	Try first
No internet	Check Wi-Fi is on, restart the router and modem, run the network troubleshooter, check if others are down too

- When you ask for help, describe it clearly: what you were doing, the exact error message, when it started, and what you already tried.
- Windows has built-in troubleshooters (Settings > Troubleshoot) for internet, sound, printers, and more.

🔤 Key vocabulary

Malware
Any harmful software

Virus
Malware that spreads through infected files

Worm
Malware that spreads by itself over networks

Trojan
Malware disguised as a useful program

Spyware
Secretly collects your information

Ransomware
Locks files until you pay

Phishing
Fake message that tries to steal your info

Social engineering
Tricking people into giving access

Firewall
Blocks unwanted network traffic

Antivirus
Finds and removes malware

HTTPS
Secure, encrypted website connection

VPN
Encrypted tunnel over the internet

Restore point
Snapshot of system settings to roll back to

Factory reset
Erase a device back to new

File History
Windows tool that backs up personal files

✅ Know for the test

- Strong passwords are long and mixed; never reuse or share them.
- Virus needs a file; worm spreads itself; Trojan pretends to be good; spyware watches; ransomware demands payment.
- Firewall blocks traffic; antivirus removes malware. Keep both on and updated.
- HTTPS and the padlock mean the connection is encrypted. Use a VPN on public Wi-Fi.
- System Restore fixes system settings but doesn't back up your files.
- Factory reset erases everything; back up first and do it before selling a device.

- Troubleshoot simple to complex: check power and cables, then restart.

🧠 Quick self-check

Answer in your head, then tap a question to see the answer.

Q1. Malware locks your photos and asks for money to unlock them. What type is it?

Answer: Ransomware.

Q2. Which spreads by itself across a network without you opening anything: a virus or a worm?

Answer: A worm.

Q3. You are on airport Wi-Fi and need to check your bank. What should you use?

Answer: A VPN (or wait for a trusted network); at least make sure the site uses HTTPS.

Q4. Will a System Restore point bring back a document you deleted?

Answer: No. It restores system settings and drivers, not personal files. Use File History or another backup.

Q5. What should you do before a factory reset of your phone?

Answer: Back up your photos, contacts, and files.

Jmal C's Digital Literacy Class · Chapter Summaries for IC3 GS5 Computing Fundamentals, Chapters 1-7. Written for this class in plain language; not a copy of the textbook.